A Machine Reads Grothendieck's Working Notes: Transcribing and Encoding Mathematical Manuscripts

Michel Hua
Commutator, Paris
michel@commutator.io

Draft of 24 October 2026, prepared for the *Journal of the Text Encoding Initiative*

Abstract

This article reports the transcription by a large language model of the working notes of the mathematician Alexandre Grothendieck (1928–2014), held by the University of Montpellier: 178 openly accessible folders, 16,074 digitised pages, 1949 to 1991, in a fast hand and a notation of his own. Twenty folders are already edited by others. The model transcribed the other 158, twenty pages at a time, into a restricted LaTeX with nine apparatus macros. A converter derives from each file a TEI document in which the mathematics is carried as TeX inside `<formula>` and commutative diagrams as `tikz-cd` inside `<figure>`. A second register, a modernised reading in current notation, is derived from the transcription and never from the hand. The edition stores no page of the fonds: the facsimile beside each reading is Montpellier's own file, passed through a relay. In all, 638 passes have read 9,414 pages in 164 folders, and the apparatus records 63,179 words left illegible, 6.4 per cent of the words met. The article sets out the encoding and its reasons, the registers of names, dates, letters and hands derived from the transcription, and the limits. No batch has been checked by a person, and only individual points have been read against the facsimile. One macro carried two senses. TEI's formula element carries mathematics without describing it. The use of generative AI is declared.

Keywords. mathematical manuscripts; large language models; transcription; critical apparatus; mathematical notation; archival rights; Alexandre Grothendieck.

1 Introduction

Alexandre Grothendieck handed his mathematical papers to his former student Jean Malgoire in two parts, in the summer of 1990 and on 28 July 1995. Malgoire kept them until 2010, deposited them at the University of Montpellier, and gave them to it in 2012 (Université de Montpellier n.d.). The university catalogued the fonds in 2015 and 2016 and has digitised part of it: 178 folders in open access, 16,074 pages by the inventory's count, in twenty-two groups in the order he filed them. They are working notes. He wrote for himself, in ink and pencil, over and between his own lines. He took the same chapter up four times, abandoned a proof and resumed it thirty pages later, and wrote on the backs of administrative letters, seminar notes and other people's typescripts.

Mathematicians have transcribed some of it by hand, over years. Their work covers twenty folders and 5,129 pages, a third of the open pages, in five works. *Les Dérivateurs* is transcribed whole into LaTeX (Künzer, Malgoire and Maltsiniotis n.d.; five folders, 2,135 pages), and so is the *Esquisse d'un programme* (Grothendieck 1997; one folder, 313 pages). Three works are

edited in part: the first thirty-seven sections of the *Longue Marche à travers la théorie de Galois* (Grothendieck 1995; four folders, 1,584 pages), the first volume of *À la poursuite des champs* (Maltsiniotis 2022; eight folders, 837 pages), and an English translation of the notes for EGA V (Grothendieck Circle n.d.; two folders, 260 pages). The correspondence with Serre (Colmez and Serre 2001) edits letters of which the folders hold a few. What he printed, EGA in the *Publications mathématiques de l'IHÉS*, now on Numdam (Grothendieck and Dieudonné 1960–1967), the SGA and the FGA, is the text some folders draft, and is not a transcription of them. The other 158 folders, 10,945 pages, had no transcription, and most of their pages had not been read since they were written.

This article reports an experiment in reading them with a large language model as the first transcriber, under constraints designed to make its reading checkable. It is the second report of a method first applied to the ledgers of Edward and Josephine Hopper (Hua 2026). Three things differ here. The document is mathematics, which is a notation as much as a language, so the encoding of mathematics in TEI becomes a central question. The corpus is two orders of magnitude larger, and the human check becomes the main fact about the edition. And the witness cannot be copied. The fonds is under copyright, a third of its pages may not be circulated, and the edition holds none of it.

The objective is methodological, and no argument here depends on a claim about Grothendieck's mathematics. As in the first report, a reading is *reproducible* when the process that made it can be rerun, *auditable* when a reader can find the evidence for each sentence, *reliable* when like cases are treated alike, and *valid* when it is right. This edition claims the first three and not the fourth.

2 Related work

The transcription of handwriting by large language models is a recent field. Humphries and colleagues (2025) report character error rates of six to seven per cent for general-purpose models on eighteenth- and nineteenth-century English manuscripts. Crosilla, Klic and Colavizza (2025) benchmark several models and discuss when a model's linguistic prior helps and when it hinders. Mathematics is an extreme case of the second. A model that has read much mathematics completes a formula as readily as a sentence, and the completed formula is well-formed, so nothing marks it as wrong. It may supply the index that makes an identity true instead of the index on the page. The edition's main rule, that nothing illegible is guessed even where the mathematics would supply it, is written against this prior, and Section 9 measures its cost.

Strutz (2026) proposes a framework for evaluating model-generated TEI, from syntactic validity to semantic accuracy. As in the first report, the model here writes no TEI. It writes a restricted language whose macros correspond one to one with TEI constructs, and the TEI is derived. The precedent for a reduced language converted into conformant TEI is Cummings and Willcox (2013), whose `tei_corset` bounded what a keying company could write. Here the macros bound what a model may write, and the converter must be testable.

The Guidelines provide `<formula>`, whose `@notation` names the notation of its content, and leave that content to another vocabulary: MathML, TeX, or an image (TEI Consortium 2026). MathML is the W3C's vocabulary for mathematical structure and presentation (W3C 2014), and a TEI file may embed it. Section 6 explains why this edition carries TeX instead, and what that choice costs. Hand attribution, which the first report treated at length, matters

less here. The fonds is in his hand, except for other people's letters and typescripts, and those are identified by what they are rather than by their letterforms (Stokes 2015).

3 The fonds

3.1 The inventory, and the three numberings

The inventory is complete and rigorous, and was made for an archive. Each folder has a shelfmark (`19`, `140–2`, `156–9`), a title, a dating and a page count. The title is his, pencilled on the cover, or the archivists', in square brackets where the folder bore none. The folders stand in twenty-two groups, from *Cristaux* and *Théorie des motifs* to the *Longue marche à travers la théorie de Galois* and *Dérivateurs*. Each folder is served as one PDF of 4 to 270 megabytes.

Three numberings overlap on every page. Grothendieck paginated some runs himself: the Long March runs from 1 to 787 across four boxes. The archivists pencilled a number at the foot of every page, and the inventory counts these; the edition calls them pages. The PDF's numbering is one ahead of the archivists', because the university's file opens on a generated cover. The edition addresses a page by the archivists' number and converts to the PDF's only when it requests a page from Montpellier.

3.2 What the pages are

The pages are drafts, and restarts are transcribed like any other page. A verso is often unrelated to its recto (an administrative letter, a typescript returned by a secretary) and was digitised deliberately, sometimes upside down. Most dates are inferred. Figure 1 sets the inventory's datings against the dates the transcriptions have found on the leaves. Of the 153 folders the inventory dates, 111 rest in part on a deduction the archivists mark with square brackets, and 25 more are « s.d. », undated. The register of dated leaves holds 252 dates in 78 folders. Seventy-nine are in his hand and 85 typed; 55 were printed by a machine, among them the banners of the computer listings whose backs he wrote on in the 1980s, and these date the paper, not the writing.

3.3 What may be read, and what was

Of the fonds' some 28,000 pages, about 18,000 may be circulated; third-party correspondence may not, without permission. The 178 folders in open access hold 16,074 pages, or 884 batches of twenty. The twenty edited folders (5,129 pages, 266 batches) are linked to their editions and not read again, apart from a few batches in six of them that the editions do not cover. Table 1 gives the state of the reading. Every batch of the 158 folders no edition covers is transcribed.

4 The transcription method

4.1 The schema, and its values here

Figure 2 gives the arrangement described in the first report. A *witness* is held by the institution and never committed. A *reading* is the one file that is written, by the model first and by a person afterwards, in a language whose every construct is declared. *Derivations* are computed from the reading by scripts on every build. *Verifications* are whatever a machine can check of them.

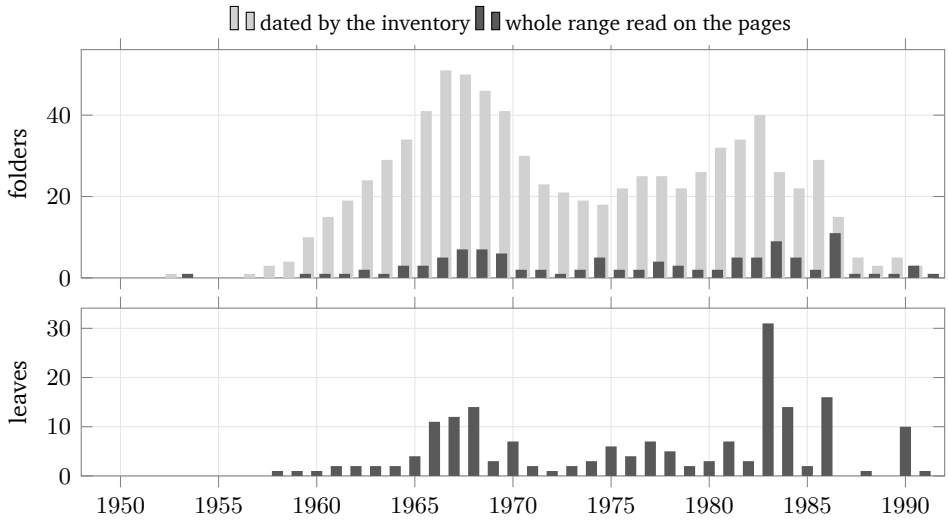


Figure 1: Two kinds of date on one axis, each on its own scale. Above: for each year, how many of the 153 dated folders the inventory's ranges cover (an open end, « à partir de 1982 », counted five years on), and, darker, how many of them rest on a range the archivists read on the pages rather than deduced in their square brackets; 25 folders are « s.d. ». Below: the 182 dates the transcriptions have found written on the leaves themselves, machine stamps on reused paper left out. From `src/content/catalogue.ts` and `src/content/dated-leaves.json` (2026-10-03), after the site's page *Timeline*.

Table 1: The fonds: the inventory, the part already edited by others and set aside, and what the passes have read. A batch is twenty of the archivists' pages; a transcribed page is one that carries mathematics, the others being skipped by rule.

	Folders	Pages	Batches
Open access (inventory)	178	16,074	884
Already edited, set aside	20	5,129	266
Left to this edition	158	10,945	618
Transcribed, folders left to this edition	158	9,129	618
Transcribed, edited folders	6	285	20
Read whole and modernised	164	9,414	638

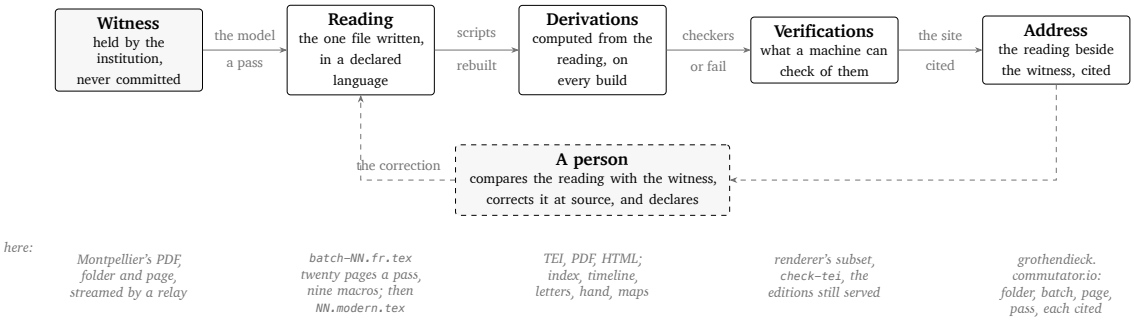


Figure 2: The schema this edition shares with its sibling on the Hopper ledgers (Hua 2026), and its values in this one. Five stations and one loop: the model turns a witness into a reading, in passes whose header names the model and the date; scripts derive everything else from the reading and regenerate it on every build; checkers fail the build on what a machine can check; the site places the reading beside the witness and cites both by reference. The dashed loop is the person, who compares the reading with the witness and corrects the reading, which alone changes anything downstream. Four invariants hold at every station: the witness is never committed; only the reading is written, everything else is derived and compared; every assertion carries an address into the witness; and the person's verdict is declared, not observed. The grey row gives this edition's values. Figure by the author.

An *address* places the reading beside its witness and cites it by the holder's reference and the edition's dated pass. The person closes the loop. What is new here is a model that also restates, a witness that cannot be mirrored for the reader, and a notation.

4.2 The unit of work and the model

A batch is twenty of the archivists' pages, read in one fresh model context and written as a single file, `transcripts/<folder>/batch-NN.fr.tex`. Beyond twenty pages of this hand, the reading degrades towards the end of the pass, and nothing in the output signals where. The skill that performs the reading admits named models only. Each file's header records the model and the date in a fixed form, `% Pass: Opus 5.5 (claude-opus-5-5), <date> --- first pass, unchecked against the pages by a human`, so that provenance is recorded in the file itself. Of the 638 first passes, 505 were read by Claude Opus 5.5, 102 by Claude Opus 5, 26 by Claude Fable 5.1 and 5 by Claude Fable 5. The Fable models are no longer used for new passes, and the headers of the files they read say so. A hundred and seventy-three files in eighty-two folders carry a later revision below the first pass, dated and attributed in the same form; most apply readings the editor settled on the facsimile, and none makes its batch checked. The batches of a folder may run in parallel. No batch then sees the one before it, and each header says where a run of text crosses its boundary.

Every pass receives two aids, and neither may settle a reading. The first is a set of tiles: each page cut into six overlapping crops at 700 dpi, with the margin turned upright, so that the pass spends no effort choosing what to look at. The second is a lexicon of his vocabulary, abbreviations and notation, counted over all transcriptions and regenerated after each run. It separates words read plainly from words read with doubt. On a hard leaf a pass usually chooses between candidates, and the lexicon shows which belongs to this material. Built from unverified transcriptions, it makes no claim about ink. It records that *principal* is common in these pages, and leaves to the page whether one stroke reads *principaux*.

4.3 The rules

The rule of the Hopper edition applies with a sharper consequence. An invented word in a letter is a bad reading, and an invented index in a formula can state a result nobody proved. Nothing illegible is guessed, and the mark for an illegible word is never replaced by what the mathematics would require. A doubtful reading is marked as doubtful. His spelling, punctuation, long sentences and slips are kept, and slips are flagged, not corrected. The transcriber's notes are in French, so that a French reader meets one language throughout.

The transcription is not diplomatic, and says so. The material apparatus of the archive (which verso carried which letter, where a separator sheet fell, which page was scanned upside down) is not its subject. A page with no mathematics is omitted, and the gap in the page numbers records it. Where a page is too crowded to render whole, the mathematics is transcribed and the surrounding prose compressed, with a note. Commutative diagrams go into `tikz-cd`, the notation in which this mathematics is read today. His prose stays whenever it concerns the mathematics, as when he explains why a definition is the right one. What he struck out stays, marked, because a deletion shows where the argument changed.

Three rules concern other people. A typescript or mémoire by someone else is given as a note per page, with his marks on it in full. A letter to him is transcribed in full where its writer is a public figure, and otherwise summarised with the writer unnamed. Letters already

published in the correspondence with Serre are condensed to their statements and a summary. Each rule answers a right the edition does not hold.

4.4 What the model writes

The model writes a restricted LaTeX. Its apparatus is nine macros (Table 2). The mathematics is ordinary LaTeX mathematics, inline and displayed, with the environments a working mathematician uses and `tikz-cd` for diagrams. Six declarations open each file: the shelfmark, the batch, its pages, the inventory's dating copied with its brackets, the folder's title, and a watermark, *Édition de démonstration*, which the PDF prints across every page.

Table 2: The apparatus macros, as the transcription skill declares them, and the TEI elements the converter maps them to. The mathematics itself is carried as TeX (Section 6).

Macro	Meaning	TEI
`\page{47}`	Page 47 begins, in the archivists' numbering	`pb` with `@n`, `@facs`
`\ill{}`	Illegible; never guessed	`gap reason="illegible"`
`\uncertain{…}`	A reading offered and flagged	`unclear`
`\supplied{…}`	What the transcriber supplies and the page does not carry: a missing letter, a clipped word-end	`supplied resp="#pass"`
`\add{…}`	His own insertion: above the line, over a struck word, his ink on a typescript	`add`
`\struck{…}`	Struck out by Grothendieck	`del`
`\note{…}`	The transcriber's note, about the mathematics	`note type="editorial"`
`\drawing{…}`	A drawing of his, described rather than redrawn	`note type="drawing"`
`\marginal{…}`	A marginal note of his	`note type="authorial" place="margin"`

Two scripts enforce the subset, the renderer of the site's reading view and the converter to TEI (Section 6), and a refusal by either stops the build. Both refuse an unknown environment. The converter also refuses a control sequence it would leave in the TEI as text, and a pass record it cannot parse. Two known faults still pass both scripts and are found only by reading the output: a backslash-newline inside inline mathematics, and a page-range declaration without a blank line before it.

4.5 A transcribed page

The following is page 8 of folder 19, *Topos*, from the first batch of that folder, as the converter exports it (the page is folder 19, batch 1). The mathematics is TeX inside `<formula>` and the diagram is a figure. The note is the transcriber's and the marginal note is his. The struck φ and its replacement are both kept.

```
<pb n="8" facs="https://grothendieck.umontpellier.fr/19.pdf#page=9"/>
<p>qui s'expliciteront en termes des <formula notation="TeX">\varphi_{ji} = f_{j} g_{i}</formula>,
     [...]
<formula notation="TeX" rend="display">\lambda_{kji} : \varphi_{ki} \longrightarrow
  \varphi_{kj} \circ \varphi_{ji} \tag{3.10}</formula></p>
<p><figure type="diagram"><formula notation="tikz-cd">\begin{tikzcd}[column sep=small, row sep=
    small]
```

```
\varphi_{ki} \arrow[r, "\lambda_{kji}"] \arrow[d, no head, "\parallel" description]
  & \varphi_{kj}\varphi_{ji} \arrow[d, no head, "\parallel" description] \\
f_{k} g_{i} & f_{k}\, g_{j} f_{j}\, g_{i}
\end{tikzcd}</formula></figure></p>
<p><note type="editorial" resp="#pass">la flèche provient du morphisme d'adjonction
<formula notation="TeX">\mathrm{id} \to g_{j} f_{j}</formula>, que l'auteur indique par une
flèche verticale sous <formula notation="TeX">g_{j} f_{j}</formula></note></p>
<p><note type="authorial" place="margin">pour <formula notation="TeX">i = j = k</formula>,
<formula notation="TeX">\lambda_{kji}</formula> n'est autre que
<formula notation="TeX">\lambda_{i}</formula></note></p>
<p>[Le cas le plus intéressant est celui où les <del><formula notation="TeX">\varphi</formula></
    del>
<add><formula notation="TeX">\alpha_{i}</formula></add> sont des
isomorphismes (i.e. les <formula notation="TeX">g_{i}</formula> pleinement fidèles). [...]</p>
```

In the last paragraph, the α_i is his, written over the φ he struck, and is encoded as `<add>`. Section 9.2 explains why an earlier mapping credited such insertions to the transcriber.

5 Two registers

The Hopper ledgers needed one text. This fonds needs two, because of the notation. A $\mathcal{C}$ of 1962 is not drawn like one of 1986, a sheaf of the Séminaire is not named as sheaves are named today, and a reader who meets « fppf » spelled out on one page and reduced to initials on the next needs to be told it is the same topology. Every folder read whole therefore yields a second document, the *modernised reading*. It opens with a résumé for a reader new to the subject, then gives the mathematics in current notation and current names, with footnotes carrying everything the transcription's apparatus carried.

Three decisions govern it. The modernised reading is derived from the transcription and never from the hand, since two independent readings of one hand would diverge with nothing to decide between them. It is made per folder, because his arguments do not stop where the batches do. And it is the one document allowed to depart from the page, so it is held to another standard. It must be correct as it stands. Where the manuscript is loose, the reading states what is true and gives what the page has in a footnote. The skill states the standard as four failure modes to check: a statement strengthened beyond what the page proves, a hypothesis dropped, a notation modernised into a different object, and a gap closed silently.

The résumé ends with a line of English keywords, the modern vocabulary to search under. They are the only part of either register written for search. The site's search, its index of subjects and its map of the mathematics are built from them (Section 7.3), and the site labels them as the reading's. The 164 folders read whole in this way carry 3,388 keywords.

6 The TEI encoding

6.1 A derived export

A script exports every transcription to TEI P5. It reads the `.tex` and nothing else, neither the facsimile nor the rendered HTML, and its mapping is that of Table 2. The reasons are those of the first report. A model asked for TEI writes it fluently, and well-formed markup with an invented attribute shows no structural sign of the error. A nine-macro language has a small surface, and each macro answers one editorial question. Because the TEI is derived, any

difference between a file and its TEI is a bug in the converter. The LaTeX is legible only to this repository; TEI can be taken in by an archive or another editor without the edition's tools.

The header carries the file's comment in machine-readable form. The responsibility statement names the model as a `<name type="model">`, which every editorial note points to. The availability is `restricted`, with the reasons: the fonds is under copyright, the transcription is an unauthorised demonstration, and no image is reproduced. The manuscript description gives the shelfmark, the inventory's title, and its dating in an `<origDate>`. The surrogate is Montpellier's PDF, with a note that each `@facs` points one page ahead of its `@n`. Every pass recorded in the source is a `<change>` in `<revisionDesc>`, dated and pointing to its model.

6.2 Mathematics as TeX

Mathematics is carried as TeX, unchanged, inside `<formula notation="TeX">`, with displayed formulas marked by `@rend`. A commutative diagram is a `<figure type="diagram">` holding a `<formula notation="tikz-cd">`. The few drawings that an argument depends on and that are not arrow graphs (arcs, chains of faces, pseudo-lines) are redrawn in TikZ and carried as `<figure type="picture">`. It holds the source in a `<formula notation="TikZ">` and, in a `<graphic>`, an SVG compiled from it, since no browser draws TikZ. Neither TEI nor the converter encodes the mathematics. There are three reasons for this, and one cost.

First, the transcription is already TeX, and TeX is closer to what he wrote than MathML would be. He wrote a notation and did not declare a structure: an underlined ω, a doubly underlined Σ, a tilde that distinguishes two coverings, one letter used for two objects on one page. Presentation MathML would be a second transcription of the first, and content MathML would be an interpretation, which belongs to the modernised reading. Second, the apparatus. An illegible exponent or a doubtful index is an apparatus mark inside a formula, written as `\ill{}` inside `$…$`. Kept as TeX, it stays where both the browser's renderer (KaTeX) and the PDF read it. A conversion to MathML would have to give an illegible subscript a structure, which nobody knows. Third, the round trip: the TeX inside the TEI is the TeX of the source, character for character, so the two can be compared as strings.

The cost is that the TEI says nothing about the mathematics that a TEI processor can use. A reader of the TEI can find every formula but cannot search inside one, and the 2,000 commutative diagrams are opaque to the schema. An apparatus mark inside a formula is invisible to TEI: a count of `<gap>` elements misses every illegible index. The edition's own counts (Section 7.3) are therefore taken from the source. MathML can be derived later from the same TeX, where the TeX is clean enough, without touching the transcription. The journal's own customisation, against which this article is encoded, admits no `<formula>`, so the article's TEI carries its few inline symbols as Unicode.

6.3 Validation

The edition has a customisation of its own, modelled on the one for the Hopper ledgers, where a corpus of well-formed files had turned out to be invalid because nothing had checked them (Hua 2026). The ODD declares the sixty-four elements the converter emits, header included. It closes the value lists of the attributes that carry meaning: `@type` on `<div>`, `<figure>` and `<note>`, `@notation` on `<formula>` (`TeX`, `tikz-cd`, `TikZ`), `@reason` on `<gap>`, and `@rend` wherever the converter writes it. It is a pure subset, so a file valid against it is valid TEI.

The RELAX NG schema is derived from the ODD on every build and published beside it, and every exported file names it in an `xml-model` processing instruction. The build validates every

file with jing against unmodified `tei_all` and against the derived schema, with every input pinned by version and digest. It also compares the ODD with the export in both directions and fails on an element used but not declared, or declared but never used. A further check typesets every formula of the TEI's reading view and of the `.tex`'s with KaTeX and compares them page by page. Two things are not validated: the Schematron rules that `tei_all` embeds, which are not run over the corpus, and the TeX inside each `<formula>`, which the schema sees as a string and only KaTeX reads.

6.4 Several witnesses of one text

Some texts survive in more than one copy, and the copies do not carry the same marks. The *Tapis de Quillen*, sixteen typed pages dated 10 September 1968, has four witnesses. In folder `162-5` of the Montpellier fonds, the typed top copy carries his corrections in blue ink, and a black photocopy taken after the ink adds marks of its own; folder 111 is another photocopy of that second state. A fourth copy, in the Fonds Grothendieck–Illusie of the IHÉS (item 2ILL3), was photocopied from the inked top copy before those later marks, and carries pencil notes written on it alone. The letter to Tate of May 1966 is in Montpellier folder 6 and, as a photocopy of the same annotated sheets taken before Montpellier's numbering, in the IHÉS item FG_IL_2_TATE66. Pages 168–185 of folder 93, on the Cartier operation, are also IHÉS item FG_IL_3_2, which keeps a corner that the Montpellier sheet has lost.

The copies were identified from internal evidence: the same typed master, with the same line breaks and the same typewriter x-outs, and a page-to-view correspondence that holds through the whole item (view N of the Tate letter is page $N + 1$ of folder 6). Each witness is transcribed in its own file, against its own images. When another witness settles a reading, the correction is made in the file it concerns, and its header records a revision that names the witness and the view it was read on. A note in the body then says which witnesses carry the mark. This is how the edition corrected some thirty readings in folders 6, 93 and 111 from the IHÉS copies, among them two marginal blocks of the *Tapis* that folder 111 had given as unreadable.

TEI offers more than this. A parallel-segmentation apparatus would declare the witnesses once in a `<listWit>` and record each variant where it occurs, as in this schematic case from leaf 3 of the *Tapis*, where a marginal "Si !" with its bracket is on three witnesses and absent from the top copy:

```
<listWit>
  <witness xml:id="B">Montpellier 162-5, typed top copy, blue ink</witness>
  <witness xml:id="A">Montpellier 162-5, black photocopy with its own marks</witness>
  <witness xml:id="M">Montpellier 111, photocopy of A's state</witness>
  <witness xml:id="I">IHÉS 2ILL3, photocopy of the inked top copy</witness>
</listWit>
[...]
<app>
  <rdg wit="#A #M #I"><note place="margin">Si !</note></rdg>
  <rdg wit="#B"/>
</app>
```

The edition does not do this yet. Each exported file encodes one witness, and the agreement between witnesses lives in prose notes, which a TEI processor cannot query. An apparatus of this kind would also need the mathematics to be segmented, since a variant often falls inside a formula, where the TeX string has no internal structure (Section 6.2).

7 Interface and reproducibility

7.1 The facsimile

The site at `https://grothendieck.commutator.io` shows the transcription in one pane and Montpellier's facsimile in the other. Scrolling the transcription past a page break turns the facsimile to that page. A reading is addressed by page: `#140-1/2/p28` opens folder `140-1` at batch 2, scrolled to page 28, with the facsimile at the same page. Every register below links in that form. The edition holds no page of the fonds, either in the repository, whose ignore file refuses PDF files, or on the server.

A frame pointed at Montpellier's server fails, for three reasons. The server sends the header `X-Frame-Options: SAMEORIGIN`; its certificate expired on 10 December 2025, and it sends no cross-origin headers. The browser enforces all three against the remote origin; none applies to a request made by a server. A relay on the edition's origin therefore requests the file and passes the bytes through unchanged, with range requests, so that opening page 221 of a 204-megabyte folder transfers 256 kilobytes. The relay serves only an allowlist of the folders' files, checked against the inventory, and stores nothing.

The reading can therefore be checked against its witness only while Montpellier's server is online. When it is down, the pane says so and offers no copy; the local mirror that the passes read from is never served.

7.2 Counts computed from the files

Everything the site shows beyond the transcription is derived from it, and every count is computed from the files. The manifest of what is transcribed is built from the files present. A batch counts as transcribed once its LaTeX exists, and as modernised once the folder's reading exists; neither status claims that anyone has checked it. The site's progress figure is computed against the folders nobody has edited, and the figure against the whole fonds is printed beside it.

7.3 Registers derived from the reading

Because the transcription is written in a constrained language, a question put to the fonds can be answered by a script. The edition publishes seven such registers. Each is regenerated from the files, and each entry rests on a file and a line.

The hand counts the apparatus per batch and per folder. Of the words met, 925,157 were read and 63,179 left illegible, 6.4 per cent; 44,520 readings were marked doubtful. He struck 39,341 times and inserted 14,722 times outside the transcriber's notes; the transcriber supplied 749 letters or words; there are 5,792 marginal notes, 2,000 commutative diagrams and 902 drawings, four of them redrawn and the others described in notes. Figure 3 places each folder's share of unread words against its inventory dating. The count is of the apparatus. An `\ill{}` is a word one pass could not read, so the count depends on the pass as well as on the hand.

The timeline draws each folder's inventory dating as a range, dark where the archivists read it and light where they deduced it, with the dates on the leaves as dots (Figure 1 summarises it). The parser keeps the qualifier of each end of a dating, « vers », « à partir de », « avant »; reducing « 1965-[vers 1971] » to 1965–1971 would lose the distinction the page records. *The letters* is a register of 93 letters in 41 folders, 1959 to 1991, each with its date as written, its direction, its form, and how the transcription gives it, private correspondents unnamed. *The*

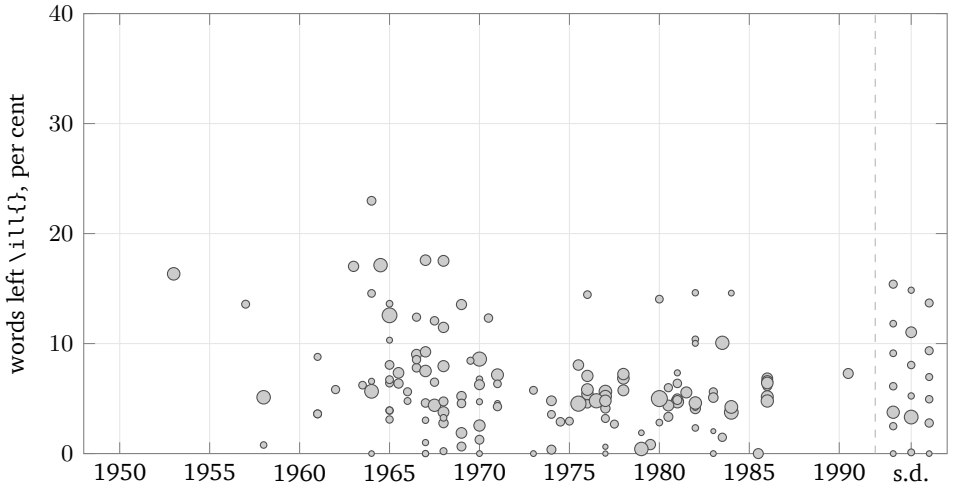


Figure 3: The share of words the first pass could not read, folder by folder, against the middle of the inventory's dating; 19 undated folders stand in the column at the right. A larger dot, more pages transcribed. 164 folders, 925,157 words read and 63,179 left illegible (6.4 per cent). No trend is drawn: the datings are ranges, often decades wide. The count is of the apparatus, not of the ink. From `src/content/hand.json` (2026-10-21), after the site's page *The hand*.

index leads from a person, a work of his cited by volume (EGA, SGA, FGA), or a subject, to the folder and batch where it occurs. *The maps* are co-occurrence networks in the manner of bibliometric mapping: who is named together in a folder, which subjects share folders, which folders cite which volumes. Every edge rests on evidence, and the layout uses a fixed seed. A slider shows each map with only the folders dated by a given year, by the midpoint of the inventory's dating.

The last two registers collect the mathematics itself. *The diagrams* are the 2,000 `tikzcd` environments of the transcriptions, in 142 folders, shown one to a slide by the reading view's renderer. *The formulas* are the 25,946 displayed formulas, in 156 folders, set by KaTeX with their apparatus. Each register opens on its richest items, ranked by a stated count: the distinct symbols a diagram or formula uses (a letter in a given alphabet, so that $\mathcal{C}$ is not C, a named symbol, an operator), plus nodes and arrows for a diagram and size for a formula. The ranking measures quantity and says nothing of importance. The richest diagram has nineteen objects and twenty-nine arrows, on page 28 of folder `140–1`, a draft of the *Longue Marche*.

Two rules govern these registers. The transcriber's notes are excluded from every count of his words, since a note discussing a notion would otherwise report it as his. A person counts as a node where the folder holds a letter to or from them, a text of theirs, or his naming them as the source of a result, as in « th. de Hodge » or « d'après Deligne ». A mathematical object that bears a name, such as « groupe de Galois », does not count. The test is adjacency in the text. A source judged unreliable is not used, even where it is the only source for a claim; the claim then stays unsourced and the graph omits it.

8 Findings

Reading a folder end to end turns up things the inventory does not record: a leaf bound backwards, two manuscripts interleaved, a theorem proved under a hypothesis weaker than the one in print. The edition keeps them in a list designed to prevent overclaiming. A claim that something in these folders is new is a claim about the literature, which can be searched

but never exhausted. A claim that it came first needs a date, which the folders mostly lack.

Each row carries a status from a closed list: *unsearched*, nobody has looked it up; *candidate*, searched and not found; *matched*, found in the literature and kept; *refuted*, shown false as stated by a counterexample, kept with what still stands; *confirmed*, which only a person may set. Every row names the sources searched and separates what the page carries from what the edition supplied. The skill that proposes rows reads the transcription and the modernised reading, never the facsimile, and may not assert priority. Matched and refuted rows stay in the list, so the next reader is spared the search. The list holds 463 rows from 155 folders: 216 unsearched, 165 candidates, 73 matched, eight refuted, and one confirmed, on the word of the author of the book it corrects. Statements of the readings of sixty-seven folders are also proved in Lean 4 against mathlib, in 2,052 theorems using only Lean's standard axioms. A proof shows that the reading's statement holds, which does not make it what the page says; in some folders it found a statement false as given.

One row shows the purpose of the list. Folder 27 holds five typed leaves headed in pencil « Exp XIII »: a proposition on a group acting on a scheme, with a chain of implications among six conditions, and three corollaries. A first pass summarised them as an already published typescript. A comparison with the exposé in the current re-edition of SGA 3 (Gille and Polo 2008–2011, Exposé XIII in its version of December 2008) showed the premise was wrong. The published exposé goes from its first lemma directly to its second section, and the general statements reappear there only in specialised form. The leaves are an earlier, longer version that is not in print, and they were re-transcribed in full. This does not establish that the version matters. It establishes that the inventory's « typescript » and the published exposé are different texts, which a reader of the folder could not have known.

9 Discussion

9.1 The reading against the image

The transcription is a first pass, and no batch has been checked by a person. The editor has checked individual points against the facsimile: dates, doubtful words, the hand of an addition, the ink of typed batches sampled one page each, untranscribed pages (found blank), the sixty-three points at which ten documents differ from a transcription made independently elsewhere, and a few pages of letters and typescripts. A hundred and fourteen batch files record such checks, each dated and signed in the file, and none makes its batch checked. The page of folder 19 in Section 4.5 was read against the facsimile in this way, and four of its readings were corrected (Appendix A). The edition's status file, in which only a person may declare a batch checked or skipped, declares no batch checked and two skipped, and the site shows every batch as drafted. The rate at which the reading fails is stated in the apparatus: 6.4 per cent of words left illegible and a further 4.5 readings in a hundred words marked doubtful, varying by folder from nothing to more than a third (Figure 3). These are not error rates. A word misread plainly leaves no mark, and only a person with the facsimile can find it.

9.2 Checking by spot: the review page

Many of these checks were made through a page of the site, `/review/`, which turns the edition's checklist into single questions. The checklist was first the body of an issue in the repository (#35) and is now kept as a file, `docs/review/issue-35.md`. A script,

scripts/review-queue.mjs, reads it with the transcriptions and builds the queue: in the batches densest in unread words it samples three \ill{} and two \uncertain{} per batch; over the corpus it samples free-standing \add{} and glued \supplied{}, one folder at a time; and it adds the dated leaves and the register of letters whose entries the checklist questions. Each item is one spot of one page. The page shows it beside Montpellier's facsimile, opened at that page, with a question in plain words (« Is the highlighted word what the page has? ») and a fixed set of answers, one of which asks for the reader's own reading in at most fifteen words.

Answers stay in the reader's browser until they are sent, and sending opens a GitHub issue already filled in, which the reader submits under their own account. The answers are then applied at source. The batch file gains a line % Revised: naming the model that applied the change, the date, the reader and the issue, and scripts/tei.mjs turns each such line into a <change> element, with @when and @who, in the <revisionDesc> of that batch's TEI header. Where a reading needs it, a \note{} on the spot says what was checked. The item is recorded in review-answered.json with its issue and a result (applied, confirmed, still illegible, still open), and the next build of the queue leaves it out.

Whether the site may show the transcriptions is not yet settled with the holders of the rights, and a switch read at build time, RENDER_TRANSCRIPTS, turns their display off. The issues are public, so they carry only the verdict and the reader's fifteen words. The current reading around each spot, also at most fifteen words, sits in a separate file that the page imports only when the build may show transcriptions; with the switch off, the file is not in the bundle.

By 23 October 2026 nineteen issues had come from the page, #61 to #65, #67 to #71, #74 to #78, #80 to #82 and #84, and settled 173 items. Seventy-four readings of words were changed or confirmed, forty of them insertions whose hand was in question; eleven words stayed illegible to the reader, and of one insertion and one doubtful word the reader could not tell. Thirty-six items of the register of letters were confirmed or corrected. Fifty-seven dates were checked on their leaves: eight were placed on their page, thirty-one confirmed or corrected in the form the page gives them, six found to be labels or titles rather than dates of writing, which the register of dated leaves now marks as such, and twelve day-only dates found to bear no year of their own, the year being written on another leaf of the folder, which the register now names; of two the reader could not tell the page. The queue now holds 10 items. A reader of the hand is not a reader of the mathematics: the items that asked whether one of the edition's own arguments holds, or whether a reference cited from memory is right, were answered « can't tell » in #69, and most of the confirmations given earlier were withdrawn. Those checks are left to specialists.

9.3 Limits of the model as a transcriber and as an encoder

There are five limits, none of them an error rate.

One macro carried two senses. \add was first declared as the transcriber's addition (a missing letter, an implied word) and mapped to <supplied resp="#pass">. In context, most of its uses were his: a word written above the line with a caret, a substitution over a struck word, the α_i of Section 4.5. The TEI thus credited several thousand of his insertions to the model. The Hopper edition had the same fault and split the macro in two, and this edition did the same. \supplied now carries the transcriber's sense and maps to <supplied resp="#pass">; \add carries his and maps to <add>. The existing uses were divided by a rule that can be checked in the file, without a new reading: an addition attached to the word it completes, with a letter

or digit immediately before or after it, is the transcriber's, and a free-standing one is his. The rule assigned 529 of the 6,144 uses to `\supplied` and left 5,615 as `\add`; later passes use the two macros from the start. The rule's output is a guess. A closing parenthesis restored at a clipped edge still reads as his, and a prefix he inserted, now « `\supplied{quasi-}torseur` », is attributed to the transcriber. Only a person reading the facsimiles can settle these cases. The general lesson is that a macro whose declared sense is narrower than its natural sense will be used in the natural one.

The model knows the mathematics. Its prior is the hazard the rule of Section 4.3 addresses, and the rule holds only where it is applied. An illegible index that the context determines is easy to fill in. The transcriptions write `\ill{}` and, where the context fixes the value, say so in a note, as in « l'indice de k est un pâté … le contexte ne le fixe pas ». A pass that supplied the index silently would leave no trace. The method separates the registers for this reason. Completion is allowed in the modernised reading, which may state what is true, and forbidden in the transcription.

The apparatus is inside the mathematics. As Section 6.2 explains, a count over the TEI misses every mark inside a formula, and a user of the TEI alone would undercount the illegible words.

Other people's texts. The fonds holds letters, typescripts and mémoires by others, some of them private persons. The rules of Section 4.3 answer to rights, and the model has to apply them. The skill judges conservatively whether a writer is a public figure, and the register of letters leaves private correspondents unnamed. Only a human reviewer can check that judgement. An error there would disclose a private person's text.

Dates. The model reads a date as readily as a word, and dates are what the edition most needs, since the inventory's datings are mostly inferred (Figure 1). The register of dated leaves distinguishes a date in his hand from a typist's and a machine's, keeps two-digit years as written, and fixes a year from a neighbouring leaf only where the run is continuous, saying so. Of its 252 records, 195 were assembled by a model reading the transcriptions and have not been checked against the facsimile.

9.4 Human validation, and the pace

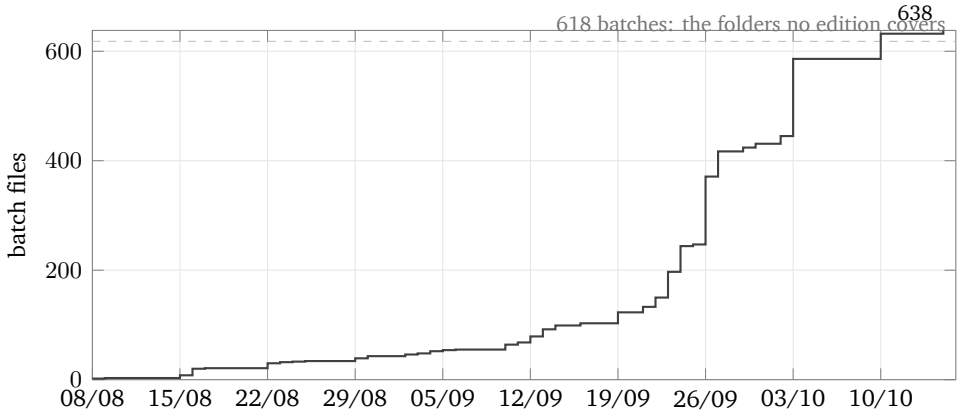


Figure 4: Batch files in the repository, day by day, from the first on 2026-08-08 to 2026-10-15: the date each `batch-NN.fr.tex` was first committed, cumulated. Each file is one pass of the model over at most twenty pages. The dashed line is the 618 batches of the folders no edition covers; the curve ends 20 above it because six edited folders also have a few batches. From the repository's git history, drawn on 2026-10-24 by `docs/study/figures.mjs`.

Figure 4 gives the pace from the repository's history. The first batch file was committed on 8 August 2026 and the 638th on 15 October: 43 in August, 388 in September and 207 in October, with up to eight passes running at once. The 618 batches of the folders no edition covers were transcribed in about eight weeks. The human check takes far longer. A mathematician who can read this hand needs about an afternoon to compare a batch of twenty pages line by line with the facsimile. The 618 batches therefore represent some two and a half thousand hours of a rare kind of reader. The checking, and no longer the transcription, limits the edition.

The protocol the edition proposes is the one its status file already encodes. A person opens a batch beside its facsimile, reads each transcribed line against the page, corrects the transcription at source with a note saying who checked it, and only then declares the batch checked. Until then every citation names a machine pass and a date, and the address of a page does not change when its reading does.

9.5 What the method permits, and what it does not

The method allows one person and a model to produce, in months, a reading of a sixteen-thousand-page mathematical archive, cited by folder, batch, page and pass, exported to TEI with its mathematics intact, placed beside the institution's facsimile without copying it, and mined for registers that each rest on a line. It does not produce a checked transcription, it does not encode the mathematics in a form a TEI processor can reason over, and it cannot decide with authority who is a public figure.

10 Conclusion

This article has described a second edition built on the method of the first, and what a corpus of mathematics changes in it. The notation required a decision the ledgers did not. Mathematics is carried inside `<formula>` as the TeX it was transcribed in, without a structure the transcription does not have, and the interpretation of the notation goes into a second register derived from the first. The size of the corpus made the human check the main fact about the edition. The rights meant that the edition points to the facsimile instead of serving it, through a relay that stores nothing.

The encoding is in place and validated on every build. Two pieces of work remain. The first is to check on the facsimile the insertions that the rule of Section 9.2 divided mechanically between him and the transcriber. The second is the human check of every batch. Every claim of the edition depends on it, and it requires mathematicians who can read this hand.

For this fonds, checking means reading Grothendieck's hand against sixteen thousand pages. Few people can do it, and nobody can do it alone. The method provides a cited reading before that check, and does not replace it.

About the author

Michel Hua is an independent consultant at Commutator, Paris, working with large organisations on data and cloud applications: platform migrations, data governance, and the adoption of generative AI. He has built editions of archives that institutions have digitised and few have read: the Hopper ledgers at the Whitney Museum, the Grothendieck fonds at Montpellier, and mathematicians' manuscripts in Gallica. `michel@commutator.io`.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude Opus 5, Claude Opus 5.5, Claude Fable 5 and Claude Fable 5.1 (Anthropic), through Claude Code, in order to transcribe the pages of the fonds into the restricted transcription language described in Section 4, to write the modernised readings described in Section 5, to write the scripts, the converter, the relay, the interface and the derived registers described in Sections 6 and 7, to propose the findings and the Lean proofs described in Section 8, and to draft this article from the transcriptions and the derived data under the author's direction. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. The transcriptions have not been checked against the facsimiles by a person, and the article says so wherever a claim depends on them.

Data availability

The scripts, the converter, the relay, the interface, the derived registers and the sources of this article are in the repository at `https://github.com/Commutator-IO/grothendieck-project`; the interface is served at `https://grothendieck.commutator.io`. The software and the derived data are released under the repository's licence, and this article under CC BY 4.0. The fonds belongs to the University of Montpellier, and the rights in the manuscripts are held by Grothendieck's heirs. The transcriptions and modernised readings are working documents for which the edition holds no authorisation; they are not published online without the heirs' authorisation, and are removed on request. No image of the fonds is stored by the edition or reproduced in this article.

References

Colmez, Pierre, and Jean-Pierre Serre, eds. 2001. *Correspondance Grothendieck–Serre*. Documents Mathématiques 2. Paris: Société Mathématique de France.

Crosilla, Giorgia, Lukas Klic, and Giovanni Colavizza. 2025. "Benchmarking Large Language Models for Handwritten Text Recognition." *Journal of Documentation* 81 (7). Preprint: arXiv:2503.15195.

Cummings, James, and Pip Willcox. 2013. "Stationers' Register Online: A Case Study of a Byte-Reduced TEI Schema for Digitization (tei_corset)." *Journal of the Text Encoding Initiative* 6. `https://doi.org/10.4000/jtei.926`.

Gille, Philippe, and Patrick Polo, eds. 2008–2011. *Schémas en groupes (SGA 3)*, re-edition. Tomes I and III, Paris: Société mathématique de France, 2011; the exposés of tome II in their versions of 2008–2009 at `https://webusers.imj-prg.fr/~patrick.polo/SGA3/`.

Grothendieck, Alexandre. 1995. *La Longue Marche à travers la théorie de Galois*, vol. 1. Edited by Jean Malgoire. Montpellier: Université Montpellier II.

Grothendieck, Alexandre. 1997. "Esquisse d'un programme." In *Geometric Galois Actions 1*, edited by Leila Schneps and Pierre Lochak, 5–48. London Mathematical Society Lecture Note Series 242. Cambridge: Cambridge University Press.

Grothendieck, Alexandre, and Jean Dieudonné. 1960–1967. “Éléments de géométrie algébrique, I–IV.” *Publications mathématiques de l’IHÉS* 4, 8, 11, 17, 20, 24, 28, 32. Digitised on Numdam, https://www.numdam.org/.

Grothendieck Circle. n.d. *Published Texts*. Edited by Leila Schneps. https://webusers.imj-prg.fr/~leila.schneps/grothendieckcircle/pubtexts.php.

Hua, Michel. 2026. “A Machine Reads the Hopper Ledgers: Constrained Transcription, Mechanical TEI Export, and Checkability as the Editorial Condition.” Under review at the Journal of the Text Encoding Initiative. https://hopper.commutator.io/article/hopper-jtei.pdf.

Humphries, Mark, Lianne C. Leddy, Quinn Downton, Meredith Legace, John McConnell, Isabella Murray, and Elizabeth Spence. 2025. “Unlocking the Archives: Using Large Language Models to Transcribe Handwritten Historical Documents.” *Historical Methods: A Journal of Quantitative and Interdisciplinary History* 58 (3). https://doi.org/10.1080/01615440.2025.2500309.

Künzer, Matthias, Jean Malgoire, and Georges Maltsiniotis, eds. n.d. *Les Dérivateurs*, by Alexandre Grothendieck. Online edition, https://webusers.imj-prg.fr/~georges.maltsiniotis/groth/Derivateurs.html.

Maltsiniotis, Georges, ed. 2022. *Pursuing Stacks (À la poursuite des champs)*, vol. I, by Alexandre Grothendieck. Documents Mathématiques 20. Paris: Société Mathématique de France.

Stokes, Peter A. 2015. “Digital Approaches to Paleography and Book History: Some Challenges, Present and Future.” *Frontiers in Digital Humanities* 2: 5. https://doi.org/10.3389/fdigh.2015.00005.

Strutz, Sabrina. 2026. “A Multi-Dimensional Evaluation Framework for Assessing LLM Performance in TEI Encoding.” *Journal of Open Humanities Data* 12 (1). https://doi.org/10.5334/johd.484.

TEI Consortium. 2026. *TEI P5: Guidelines for Electronic Text Encoding and Interchange*. https://tei-c.org/guidelines/p5/.

Université de Montpellier. n.d. *Archives Grothendieck*. https://grothendieck.umontpellier.fr/.

W3C. 2014. *Mathematical Markup Language (MathML) Version 3.0*, 2nd ed. W3C Recommendation, 10 April 2014. https://www.w3.org/TR/MathML3/.

A A complete transcribed page

Page 8 of folder 19, batch 1, as the model wrote it in the restricted language, in the first pass, by Claude Opus 5; the TEI of Section 4.5 is derived from the corrected file. The page is on the site, beside Montpellier’s facsimile.

```
\page{8}
qui s'expliciteront en termes des $\varphi_{ji} = f_{j} g_{i}$, et des
$\bar{\alpha}_{i}$, $\bar{\lambda}_{i}$ correspondants, ainsi
\[
  \left\{
```

```
  \begin{array}{l}
    \bar{\alpha}_{k} = \alpha_{k}(X_{k}) \circ \mathrm{pr}_{k} \\[2pt]
    \bar{\lambda}_{k,j,i} = \lambda_{kji}(X_{i}) \circ \mathrm{pr}_{i}
  \end{array}
  \right.
  \tag{3.9}
\]
\[
  \lambda_{kji} : \varphi_{ki} \longrightarrow \varphi_{kj} \circ \varphi_{ji}
  \tag{3.10}
\]

\begin{tikzcd}[column sep=small, row sep=small, nodes={font=\scriptsize}]
\varphi_{ki} \arrow[r, "\lambda_{kji}"] \arrow[d, no head] & \varphi_{kj}\varphi_{ji} \arrow[d, no
     head] \\
f_{k} g_{i} & f_{k}\, g_{j} f_{j}\, g_{i}
\end{tikzcd}

\note{la flèche provient du morphisme d'adjonction $\mathrm{id} \to g_{j} f_{j}$,
que l'auteur indique par une flèche verticale sous $g_{j} f_{j}$}

Les $(\varphi_{ji}, \lambda_{kji})$ sont reliées par des conditions exprimant
(1.5), (1.6), que nous n'expliciterons pas.

\marginal{pour $i = j = k$, $\lambda_{kji}$ n'est autre que $\lambda_{i}$}

[Le cas le plus intéressant est celui où les \struck{$\varphi$}
\add{$\alpha_{i}$} sont des isomorphismes (i.e. les $g_{i}$ pleinement
fidèles). Dans ce cas, \struck{\ill{}} on peut supposer
$\varphi_{i} = \mathrm{id}_{B_{i}}$,
$\alpha_{i} = \mathrm{id}_{\mathrm{id}_{B_{i}}}$, $\lambda_{i} = \mathrm{id}$.]

Les conditions (1.) \struck{les} dans les cas où deux des $i, j, k$ sont égaux,
avec $i = j$ ou $j = k$, \uncertain{ne montrent que} l'un des $\lambda_{kji}$
est l'identité, \ill{} de $\varphi_{kj} \varphi_{ji}$.
\note{ligne très surchargée ; le verbe reste douteux}

Il reste donc à examiner les cas (3.11) : a) \struck{$i$, $j$, $k$ deux à deux
différents} ; b) $i = k \neq j$, d'où
\[
  \mathrm{id}_{B_{i}} \longrightarrow \varphi_{ij} \circ \varphi_{ji}
  \qquad (i \neq j)
\]
```

Read against the facsimile by the editor, the page needed four corrections. The two vertical strokes of the boxed diagram (3.10) are equality signs, which the pass had drawn as plain lines. Case a) of (3.11) reads « i, j, k tous différents » and is not struck; the pass had read it as struck and as « deux à deux différents ». On a second reading (15 October 2026), the struck words after « Dans ce cas, » are « λ_{i} est », and the line on the conditions reads « deux des indices i, j, k consécutifs » and, in its right order, « dès lors que l'un des deux facteurs de φ□□φ□ est l'identité ». The file carries all four corrections and two `Revised` lines naming them. The listing above is the first pass.